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CHM 222 — Past Questions & Materials
by Renance

Course: CHM 222
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 3
Levels: 200

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Paper 1 of 3 — CHM 222 Second Semester Past Question 202302024

Level 200 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN, NIGERIA
COLLEGE OF SCIENCE
DEPARTMENT OF CHEMISTRY
SECOND SEMESTER EXAMINATIONS
CHM 222: STRUCTURE AND BONDING
INSTRUCTIONS: ANSWER ALL QUESTIONS
UNIT:2 CREDITS TIME:2HRS

QUESTION ONE

- (a) Give the formulae for calculating Formal charges (5 mrks) ✓
 (b) Using dots (.) and crosses(X) to represent valence electrons draw Lewis structure for
 i. PBr_3 ii. NO_2^- Assign oxidation numbers to each atom (7 mrks) ✓
 (c) Construct a molecular orbital energy level diagram for HCl molecule (5.5 mrks).

QUESTION TWO

- (a). List the major limitations of the octet rule (5.5 mrks) ✓
 (b) For each of these bonds: $\text{H}-\text{Cl}$ ✓ $\text{C}-\text{O}$ ✓ $\text{N}-\text{O}$ ✓
 i. Show which atom is more electronegative using delta notation (δ^+ and δ^-), and
 ii. Draw an arrow from the less electronegative atom to the more electronegative atom
 (6mrks).
 (c) Use the molecular orbital theory diagram to predict the bond order and number of unpaired electrons in He_2^+ and H_2^+ molecules (6 mrks) ✓

QUESTION THREE

- a. Using electron dot symbols and VSEPR theory, show and predict the structures and shapes of the following: SF_6 , CH_4 , SO_3 , BrF_3 , XeF_4 (8 mrks) ✓
 b. Determine the number of lone pairs and bond pairs in NH_3 , SO_2 , H_2O , CO_2 and ClF_3 (5 mrks)
 c. Give the shapes of s, p and d orbitals (4.5 mrks)

QUESTION FOUR

- a. Give three rules necessary in filling electrons in the molecular orbitals (6 mrks) ✓
 b. What do you understand by degeneracy of molecular orbitals (with examples) (4 mrks) ✓
 c. Be_2 exists, True or False? Explain (7.5 mrks) ✓

END OF PAPER LECTURERS DR J.I.MBONU & DR E.E.ELEMIKE

Paper 2 of 3 — CHM 222 Structure and Bonding 2023/2024

Level 200 • Second Semester • PastQuestion

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DEPARTMENT OF CHEMISTRY

COLLEGE OF SCIENCE FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

Academic Session: 2023/2024, Course title: Structure and Bonding, Course Code: CHM 222

Instruction: Answer any 3 questions. Time: 2 HRS

1. (a). Explain the following: (i). Valence Bond Theory

Hybridization

(b). Explain with example(s) the different types of Hybridization you know.

2. (a). Calculate the screening effect and Z_{eff} for 4s electron in (i) Mn ($Z = 25$) (ii) Cu ($Z = 29$) (iii).

(iv). Molecular Orbital Theory (iii)

K ($Z = 19$).

(b) Identify the four quantum numbers for the last valence electrons in calcium and potassium atoms.

(c). Calculate the standard enthalpy change for a reaction.

$CO_2(g) + H_2O(l) \rightarrow CO(g) + H_2C(g)$

Given that $\Delta_f H^\circ$ for $CO(g)$, $CO_2(g)$ and $H_2O(l)$ as -393.5, -393.5 and -285.8 kJ/mol respectively.

or (a), Write short notes on the following:

(i) homonuclear diatomic molecules (ii) Heteronuclear diatomic molecules.

(b). Describe the applications of molecular orbital theory in (i) Homonuclear and

(ii) Heteronuclear diatomic molecules.

4. (a). Describe the term ionic bond

(b). Mention three (3) characteristics of ionic bond.

(c). Define the following terms (i) Lattice Energy (ii) Dissociation Energy

(iii). Atomization Energy.

(d). Use the Born-Haber cycle to calculate the lattice energy of NaCl.

Given that: 1st Ionization energy for Na = 495.5 kJ/mol.

2nd Ionization energy for Na = 4560 kJ/mol. Electron affinity for Na = 53 kJ/mol.

Electron affinity for Cl = 349 kJ/mol. Energy to dissociate 4 mol of Cl_2 into Cl atoms = 432 kJ/mol.

Lattice Enthalpy (Na) = 108 kJ/mol.

$\Delta_f H^\circ$ (NaCl) = -411 kJ/mol.

5. (a). State the Coulomb's law?

(b). Mention the factors affecting Lattice enthalpy?

(c). State the concepts the Born-Harber cycle uses to determine the lattice energy of an ionic solid. (d) Use the Born-Harber cycle to calculate the lattice energy of MgO.

Given that: 1st ionization energy for Mg = 738.1 kJ/mol. 2nd ionization energy for Mg = 1450 kJ/mol. Electron affinity O = -141 kJ/mol. Electron affinity for O⁻ = +789 kJ/mol.

Energy to dissociate 4 mol of O₂ into O atoms = 498.7 kJ/mol. Latent heat of sublimation (Mg) = 130 kJ/mol. ΔH_f° (MgO) = -601.8 kJ/mol

Paper 3 of 3 — CHM 222, 2024 past question

Level 200 • Second Semester • PastQuestion

DEPARTMENT OF CHEMIST

COLLEGE OF SCIENCE FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

Academic Session: 2023/2024, Course title: Structure and Bonding, Course Code: CHM 222

Instruction: Answer any 3 questions. Time: 2 HRS

1. (a). Explain the following: (i). Valence Bond Theory

Hybridization

(b). Explain with example (s) the different types of hybridization

(c). Molecular Orbital Theory (iii)

that you know.

2. (a). Calculate the screening effect and Z_{eff} for 4s electron in (i) Ga ($Z = 29$) (ii) Cu ($Z = 29$) (iii). K ($Z = 19$).

(b) Identify the four quantum numbers for the last valence electrons in calcium and potassium atoms.

(c). Calculate the standard enthalpy change for a reaction,
 $CO(g) + H_2O(g) \rightleftharpoons C(s) + H_2(g)$

_ Given that ΔH_f° for $CO(g)$, $CO_2(g)$ and $H_2O(g)$ as -393.5, -393.5 and -241.8

kJ/mol. Write short notes on the following:

(i) ionic and nuclear diatomic molecules (ii) Heteronuclear diatomic molecules.

(b). Describe the 'applications of molecular orbital theory

in (i) Homonuclear and

(ii) Heteronuclear diatomic molecules

4. (a). Describe the term ionic bond

(b). Mention three (3) characteristics of ionic bond,

(c). Define the following terms (i) Lattice Energy (ii) Dissociation Energy

(iii). Atomization Energy.

(d). Use the Born-Haber cycle to calculate the lattice energy of NaCl.

Given that: 1st Ionization energy for Na = 495.5 kJ/mol.

2nd Ionization energy for Na = 4,560 kJ/mol. Electron affinity for Na = 52

kJ/mol. Electron affinity for Cl = 349 kJ/mol. Energy to dissociate 1 mol of Cl_2 into Cl atoms = 242 kJ/mol. Heat of hydration (Na) = -405 kJ/mol.

ΔH_f° (NaCl) = -411 kJ/mol.

(a). State the Coulomb's law?

(b). Mention the factors affecting Lattice enthalpy?

(c). State the concepts the Born-Haber cycle uses to determine the lattice energy

of a solid. (d) Use the Born-Haber cycle to calculate the lattice energy of MgO.

Given that: 1st Ionization energy for Mg = 738 kJ/mol. 2nd Ionization energy for Mg = 1450 kJ/mol. — Electron affinity O = -141 kJ/mol. Electron affinity for O⁻ = +789 kJ/mol. Energy to dissociate 1 mol of O₂ into O atoms = 498.7 kJ/mol. 4 H sublimation (Mg) = 136 kJ/mol. AA Hy! (MgO) = -601.8 kJ/mol 1.8 kJ/mol respectively. of an ionic